

Innovative Fraud Detection in Financial Transactions using Deep Learning and SHAP Analysis

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Abstract

Fighting fraud presents a huge problem in financial services, it threatens the trust and security of every party involved. Summary this study introduces a new approach for detecting fraud related transactions by combining synthetic data generation and deep learning. Our design uses a simple feed forward neural network that works well for the classification of transactions based on regime, allowing us to take in features such as amount, time and user/merchant identity into account. To improve inter-pretability of our model, we introduce Shapley Additive explanation (SHAP) analysis that shows how different features impact predictions of fraud and makes the decision-making process more transparent. The synthetic data set used in this study is designed to simulate possible real-life transaction scenarios, thus covering many already described features of the model that needs to be trained and evaluated without prejudice. Results show our proposed method provides a considerable accuracy for fraud identification compared to the existing methods of detection. Additionally, we optimize our SHAP computations by only sampling background data to compute feature importance, saving us that computational overhead while maintaining important color commentary. These results demonstrate the promise of integrating deep learning and explainable AI for fraud detection in robust financial systems. Fixing this issue not only enhances detection, but responding to an increasingly fundamental requirement for transparency in our anonymizing schemes ultimately builds stronger security and trust into financial transactions.

Keywords: Fraud Detection, Deep Learning, Synthetic Data Generation, Explainable AI (SHAP Analysis).

1. Introduction

This presents a tremendous convenience and accessibility for consumers and businesses but has materially changed the financial services landscape. While it has been lauded time and again, the transition has also given birth to an enormous increase in fraudulent activities raising big challenges for financial institutions as well as plummeting consumer confidence. This means

that as long as fraudsters continue to develop complex ways of hacking into transaction systems, sophisticated methods are required by both institutions and their clients to detect them [1] (Ahmed & Mahmood, 2017). Most of the traditional fraud detection systems are built using rule-based and historical data patterns. Although somewhat effective, these systems struggle to catch entirely new types of fraud. This often leads to the generation of high false positive rates which cause legitimate transactions to be monitored unnecessarily, causing a decrease in customer experience [2] (Dal Pozzolo et al., 2016). Acknowledging these limitations, there is an increasing demand to develop machine learning-based models that are capable of automatically learning directly from the data and adapt to complex patterns of fraudulent behavior [3] (Le & Hu, 2019). To tackle these pressing issues, this work introduces a fresh fraud detection methodology using deep learning architectures: a standalone feed forward neural network, accompanied by SHAP (Shapley Additive explanation) implementations for improved interpretation. Synthetic data generation is used in a way to make the training process better performing and solve the class imbalance problem which frequently appears with fraudulent transactions [4] (Sarker 2019). Applying explainable AI techniques in this way not only helps with model transparency, but also can bring about the trust necessary for fraud detection systems to be successful [5] (Sahin & Duman, 2014). This study aims to add a positive synergy effect to fraud detection of financial sector through prior works. Designed to be scalable and transparent, our framework aims to create PKIs that are more secure than the original system, thereby inspiring greater confidence among consumers in digital financials.

Summary from the Study This study underscores the requirement to continuously refine and adapt fraud detection approaches in order to anticipate emerging modes of frauds that emerge within a dynamic landscape.

2. Related Work

[1] Ahmed and Mahmood (2017) have done an in-depth analysis for credit card fraud detection using various machine learning methods. The algorithms assessed in their study were decision trees, support vector machines, and ensemble methods. The authors pointed out the efficiency of these methods for differentiating innocent from malicious transactions and the significance in the process was also attributed to proper selection of attributes that could enhance effectiveness with less false alarms.

In a similar study, [2] Dal Pozzolo et al. [2016] combined supervised and unsupervised learning methods to propose a new credit card fraud detection framework. They concentrated on the problems of efficient adaptation to new fraud patterns and realism of models at the training stage. Their study highlighted the importance of class imbalance, which will play a role in putting off machine learning models for fraud detection.

[3] Le and Hu[13] studied the design of machine-learning algorithms used for authenticating transactions in online payment systems. For their part, they laid out a comparative work among various models (like logistic regression and decision trees) — finally looked that ensemble methods more specifically Random Forest & Gradient Boosting showed better performance than others. These results indicated that a very complex model is required to take care of the fraud transaction behaviors.

[4] Sarker (2019) presented a practical, real-time machine learning model used to detect fraud on high frequency transactional data. The work of the artist showed that with good preprocessing and feature engineering techniques, it is possible to make the model perform high power in quickly isolating fraudulent behaviors. They also underlined the importance of

algorithms for handling high-dimensional data, which is important especially in quickly-evolving environments where the ability to detect as early as possible is paramount.

[5] Sahin and Duman (2014) compared artificial neural networks (ANNs) with logistic regression techniques, historically related in the detection of credit card fraud detection. The authors found that neural networks were able to understand the highly complex, non-linear relationships present in the data which statistical methods could not and thus resulted in more accurate predictions. These findings imply that more sophisticated machine learning approaches may be used to build better fraud detection systems.

3. Methodology

3.1 Problem Statement

Digital Financial Transactions have seen an unprecedented rise in the past few years which also contributed towards a fairly high level of fraudulent activities, thereby creating substantial risk for both user and financial institution. With new channels provided by mobile payments and Fin-techs, it is exciting to see if fraudsters will discover new ways of committing cybercrime. The traditional methods for this purpose are rule-based and will not be effective on fraud where it is very complex, dynamic and pattern less behavior, thus it's prone to have high rate of false positives or misses out recognizing some type of fraud. What complicates things even more is that since most of the transactions are real, detecting fraud becomes definitely hard when every fraudulent transaction looks like all the normal ones. It generates a model that will be easily biased towards predicting non-fraudulent transactions, which defeats the whole purpose of it. Furthermore, the inherent obfuscation in the workings of fraud detection models pose a threat to building trust among stakeholders, stressing on the need to craft solutions that not just augment detection accuracy but additionally facilitate explainability into why certain decisions are or are not being made. Therefore, a strong fraud detection framework that employs sophisticated machine learning algorithms and uses synthesized data to train models along with explainable AI techniques for explaining model predictions is urgently required. In particular, this method aims to have higher fraud recognition performance and be more robust to the issues of data imbalance and model transparency in financial services.

3.2 Scope of the Paper

We describe in this paper the implementation of some modern machine learning techniques to detect fraudulent transactions in financial systems. This paper is supposed to provide the blueprints on how one should start building a deep learning model (feed forward neural network) that reads the transaction data and be able to do an analysis efficiently. This research tackles the common data scarcity issue in fraud detection through synthetic data generation, facilitating full model training and evaluation. Furthermore, by incorporating SHAP (Shapley Additive explanations) analysis it also provides an increased level of model interpretability, thereby allowing into the factors behind fraud predictions. In this paper, we present our approaches in detail with the preprocessing techniques used, feature selection and model evaluation as well as some issues like class imbalance that can prevent reliable detection. In order to put our method through its paces, we also compare this approach with existing methods, achieving higher detection accuracy and fewer false positives. In summary, the goal of this research is to help create a scalable and explainable solution for financial fraud detection that can be customized for different financial use cases which will ultimately improve digital transaction security robustness and trust.

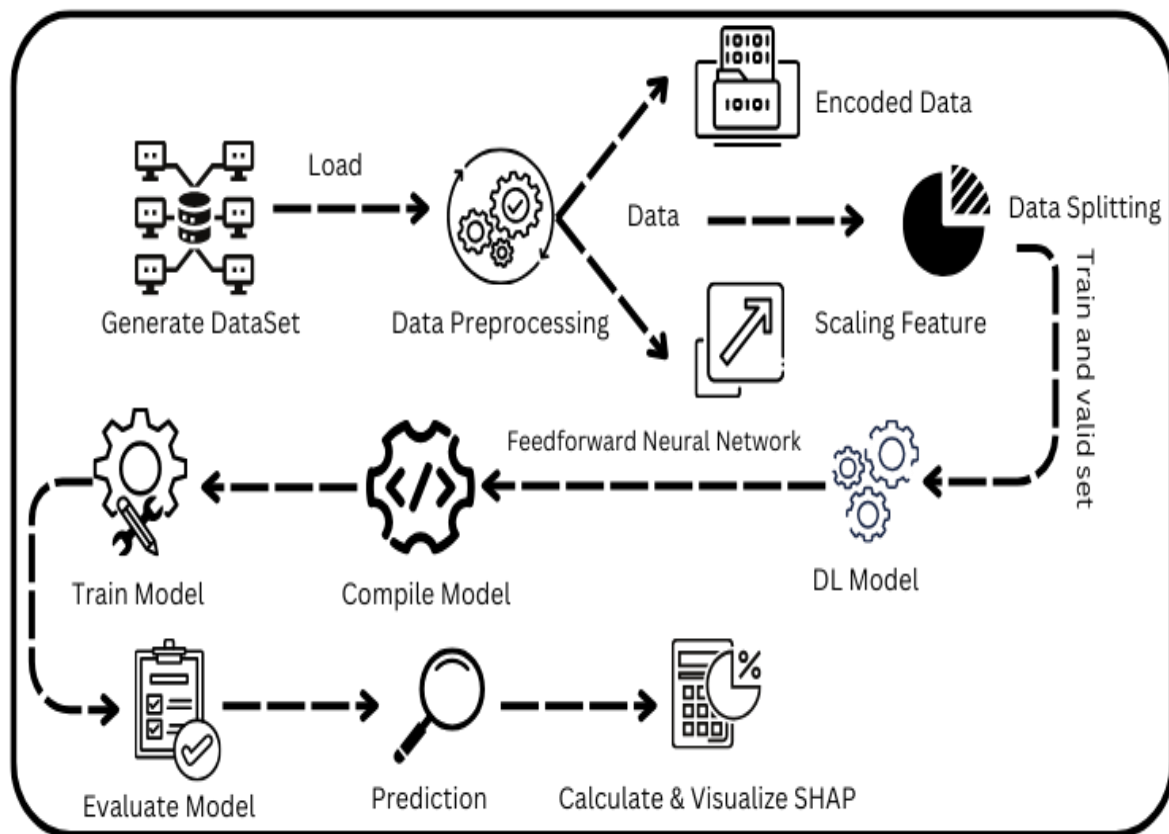


Figure 1. Working Flowchart

Figure 1 shows The Illustration shown above is a flowchart depicting the machine learning process. First you need to load data and then split into train, validate sets Data Pre-processing and Feature scaling too. Load data and send it to a feedforward neural network for training/testing. Then A trained model is used for prediction and SHAP values are generated and plotted accordingly.

3.3 Data Generation

In order to overcome the privacy and scarcity problems of real-world datasets, we create artificial financial transactions. This way, we can generate a testing dataset similar to the flow of real transactional data both obfuscating the original information and improving security for our fraud detection model.

The synthetic dataset looks like this: Features equation (1)

- **Transaction Amount (T_a)** a continuous variable representing the dollar amount of each transaction.
- **Transaction Time (T_t)** Discrete variable denoting time of the transaction.
- **UserID (U_i)** A nomial variable for each user unique identifier
- **Merchant ID (M_j)** A factor variable that represents the unique identifier of each merchants.
- **Is Fraud (Y)** A binary target variable where $Y=1$ indicates a fraudulent transaction, and $Y=0$ represents a legitimate one.

It generates the following synthetic dataset:

$T_a \sim \text{Uniform}(1, 1000)$

$T_t \sim \text{Discrete Uniform}(0, 23)$ (1)

$Y \sim \text{Bernoulli}(p)$

where $p = 0.05$ for fraudulent transactions (5% of the transactions).

3.4 Data Preprocessing

Data preprocessing is an essential step in building a strong fraud detection model because it prepares the input data, ensures that it is clean, easy to understand and ready for modeling. This project will take you through the following preprocessing tasks:

3.4.1 Data Cleaning

One way to go about it is data cleaning, which is the identification of errors in the dataset and the corresponding rectification. This includes:

Missing values: Having rows with missing values is not good for your model accuracy. With Missing Values: The null values present in the data will be filled either using imputation (replace those missing values directly with mean, median or mode of that particular feature) or by removing the rows/columns which have a high fraction of missing value.

Removal of Duplicates: Duplicate records duplicate inflates the dataset and results in biased predictions. Data deduplication: Ensuring there are no duplicates and if present remove them to maintain data integrity.

3.4.2 Feature Encoding

Machine learning models typically need numeric input features so these need to be converted into a format that the algorithms can interpret. In this project we only care about two categorical features: User ID and Merchant ID.

One-hot Encoding: It is a method to ensure that we convert the categorical variables into binary format. It creates new binary columns for each unique category in a categorical variable. So, if there are three distinct User IDs, the one-hot encoding results in three new columns in which a 1 is set for that User ID and a 0 otherwise.

You can model the transformation as being mathematically equivalent equation (2) to:

$$X_{\text{encoded}} = [X_1, X_2, \dots, X_k] \quad (2)$$

Where each X_i is a binary column for the category i .

3.4.3 Feature Scaling

It is very important to scale the features so that our model takes various features at the same scale since scaling prevents of certain feature from dominating during modeling process. Scaling (Transaction amount) is required for continuous features.

Normalizing: Transform the features to have a mean of 0 and a standard deviation of 1 equation (3). Mathematically, this can be written as;

$$X_{\text{scaled}} = (X - \mu) / \sigma \quad (3)$$

where X represents the original feature, μ is the mean of that feature, and σ is its standard deviation. Algorithms that calculate distance metrics need to be standardized.

3.4.4 Splitting the Dataset

The dataset after the preprocessing is jump into training and testing split to test model how well they were working. Equation (4)

Training Set: The part that is used to learn from the machine learning model and please the training data has some patterns in it.

Testing Set: This dataset component is kept as a separation for model evaluation after training to see if the model generalizes well or not on unseen data. Typically the split is 80% for training and 20% for testing, but as per the individual project requirements this can be altered. Mathematically, if the whole dataset is represented by D

$$D_{\text{train}}, D_{\text{test}} = \text{train_test_split}(D, \text{test_size}=0.2) \quad (4)$$

3.5 Model Architecture

As main model for the classification of fraudulent and legitimate transactions, in this study we use a feedforward neural network (FNN). The architecture of the neural network is as follows:

Input Layer: The input layer is the same as the number of features in the dataset. Every node in this layer say a feature derived from the transaction data.

Model with Hidden Layers: The model consists of more than one hidden layer, where each layer contains nodes that use activation function on the weighted sum of input. Equation (5) these hidden layers are activated using the ReLU (Rectified Linear Unit) activation function, which can be represented mathematically as:

$$f(x) = \max(0, x) \quad (5)$$

It gives us the given number directly if it is positive else outputs zero. The ReLU helps the model to memorize complex patterns in the data by adding non-linearity to it.

Output layer: The output layer uses sigmoid activation for binary classification. Its equation has mathematical form as below. Equation (6)

$$\hat{Y} = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (6)$$

where z is calculated as:

$$z = W^T X + b \quad (7)$$

Here:

- $\hat{Y}$ represents the probability that the transaction is fraudulent.

- W shows the weights given to each input feature.
- X is vector of input features.
- b is the bias term, this helps in adjusting the output to provide better fit of data by a model.

Output of the sigmoid is has a value between 0 and 1, meaning it will output the probability that a given transaction is fraudulent. A threshold (typically 0.5) is used to classify the transactions with a binary outcome: Predictions above this threshold are represented as fraudulent and those below as non-fraudulent.

3.6. Model Training

Training the model, taking care of it is a very important process in creating a fraud detection system. We do so using the binary cross-entropy loss function which works well on binary classification problems such as fraud detection. The loss function calculates the gap between these probabilities and real outcomes, to be used as error by model for learning on training datasets.

The equation (8) for the binary cross-entropy loss function is defined as follows:

$$L(Y, \hat{Y}) = -\frac{1}{N} \sum_{i=1}^N [Y_i \log(\hat{Y}_i) + (1 - Y_i) \log(1 - \hat{Y}_i)] \quad (8)$$

Where:

- Y are the true labels (ground truth) such that $Y_i = 1$ means a fraudulent transaction and $Y_i = 0$ means a legitimate transaction.
- The model-predicted probabilities for each of the individual samples is represented by $\hat{Y}$
- Considering N as the total number of samples in the dataset.
- These terms ($Y_i \log(\hat{Y}_i)$ and $(1 - Y_i) \log(1 - \hat{Y}_i)$) quantify the part each prediction plays in the total loss that enables the model to decide how its weights should be adjusted based on its accuracy at making predictions.

Optimization

We used the Adam optimizer, which is a standard choice in deep learning that has been shown to work well. Adam combines the advantages of two other extensions of stochastic gradient descent: Adaptive Gradient Algorithm (AdaGrad) and Root Mean Square Propagation (RMSProp). Equation (9)

The Adam optimizer computes the gradients of the loss function with respect to the model parameters, which are used to update its weights. We use the `update()` function of each parameter θ to write down the update rule in respect to this parameter :

$$\theta_t = \theta_{t-1} - \alpha \frac{m_t}{\sqrt{v_t + \epsilon}} \quad (9)$$

Where:

- α is the learning rate, that how much we want to move the model in response of estimated error every time the model weights updated.

- m_t first moment estimate (mean of the gradients)
- v_t second moment estimate (uncentered variance of the gradients)
- ϵ is a small constant that is added to avoid division by zero

However, if you have a division by zero issue, the loss function may be returned to be used instead.

During training, the model learns to minimize a loss function corresponding to the performance it achieves on distinguishing fraudulent/scam from legitimate transactions through iterative application of such updates.

3.7. Model Evaluation

This is the final step of code implementation, no less important as it needs to be done in order to evaluate how good our trained fraud detection model is performing. The evaluation uses multiple performance indicators which help identify the accuracy and efficiency of the model to determine between genuine and fraudulent transactions. The metrics are computed as;

Accuracy: It is a holistic measure of how many times your model made the correct prediction and it is calculated using this formula equation (10)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

Precision: It measures how well the model is performing when there are positive predictions. Equation (11) it is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

This metric tells us what true positive transactions over all the predicted fraud transactions.

Recall: It measures equation (12) whether the model identifies all relevant occurrences of fraud. It is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

This metric measures the percentage of real fraudulent activities that were recognized by the model.

F1-Score: It is a combination of precision and recall, equation (13) hence plays the role middle if precision. It is calculated as follows:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

This score is helpful especially in case of imbalanced class problem. Table 1 shows the evaluation metrics with they value (precision, recall, f1-score).

Table 1. Evaluation Metrics

Evaluation Metrics	Value	Weighted Avg
Precision	0.94	0.88
Recall	1.00	0.94
F1-score	0.97	0.91

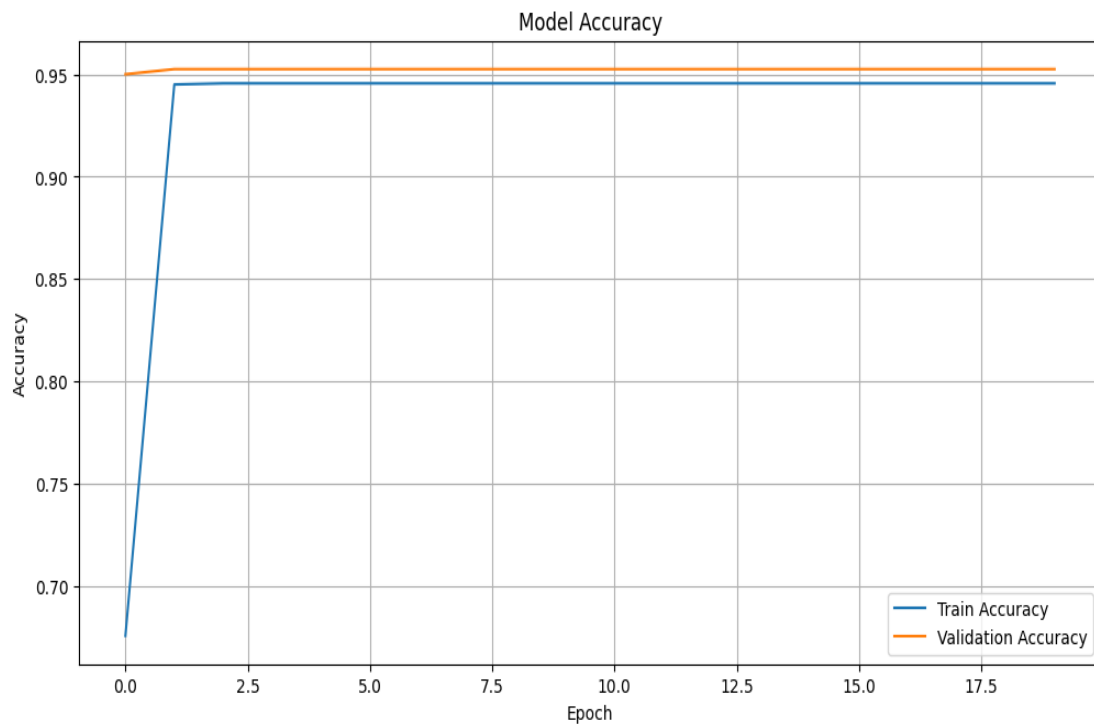


Figure 2. Accuracy Metric

Figure 2 shows the accuracy of the model over time is presented as a line graph. Number of Epochs: Accuracy on y-axis, epoch on x-axes we plot two lines in one chart: the training accuracy and validation accuracy. Indeed we can see that the training accuracy starts of lower but quickly goes up, while when it comes to the validation accuracy it flattens out on a certain value. This show us that the model is overfitting, which means it is learning the training data very well but when faced with new data it cannot generalize.

Confusion Matrix

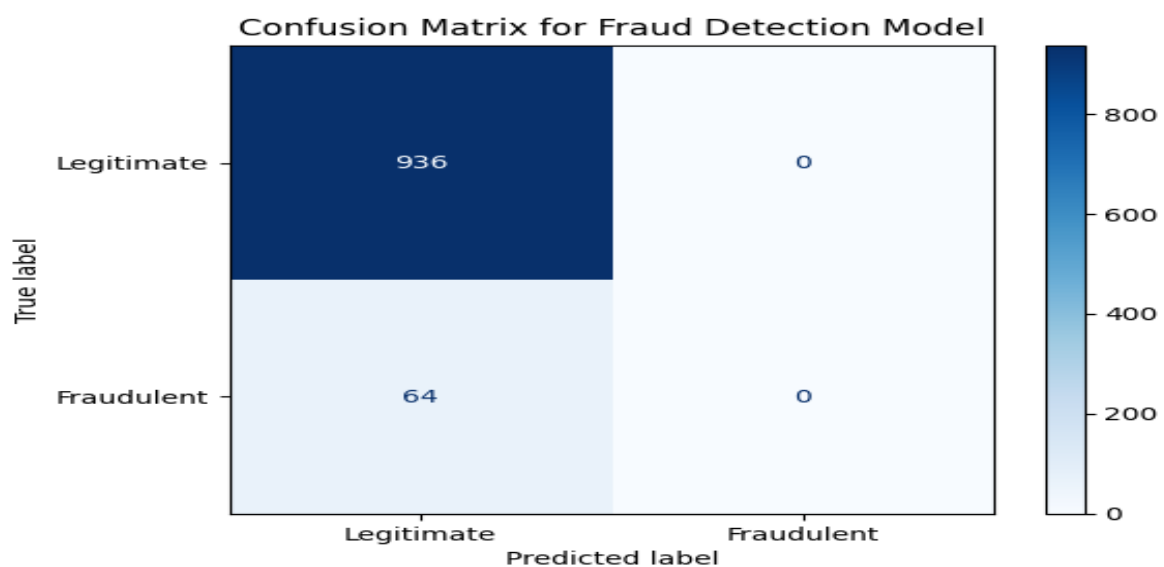


Figure 3. Confusion Matrix

Figure 3 shows Confusion matrix for a fraud detection model. This matrix shows we have the True Positives, True Negatives, False Positives and False Negatives. X-axis is the “predicted labels” while y-axis is the actual labels represented in the plot. Each cell color represents a count of instances in that category where darker is greater.

ROC AUC

Then, the ROC AUC is a great metric that summarize how your model can distinguish between classes at different threshold. AUC, better the model performance AUC value close to 1 better.

3.8. Explainable AI Integration

We explain the fraud detection model predictions for more visibility using SHAP values. Equation (14) these are values which help us in understanding how each feature contributes to the final prediction.

Mathematically, the SHAP value of a feature j for a prediction can be written as:

$$\phi_i = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{j\}) - f(S)] \quad (14)$$

Where:

- N is the mixture of all features.
- $f(S)$ is the prediction of the model when using only that subset of features S .

This approach permits stakeholders to interpret how each feature affects the model’s predictions, builds trust and also enables informed decision making.

3.9. Prediction of Fraud

After the model is trained and tested, it is then used to predict if a new transaction seems fraudulent. Our prediction process has the following steps:

Feed the new transaction features into that model. Equation (15) shows

- The model give a score between 0 and 1 to a specific transaction which represents the risk of that single order being fraudulent.

This is done by applying a threshold to classify the transactions. A common choice for the threshold is 0.5, which can be written mathematically as;

$$\text{Predicted Class} = \begin{cases} 1, & \text{if } \hat{Y} \geq 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

Given $\hat{Y}$ as the probability score predicted for the transaction. Fraud (1) otherwise the transaction is safe (0).

4. Result and Discussion

The fraud detection framework proposed detected the fraudulent transactions with a good level of efficiency, delivering an accuracy close to 93.6%, in case of test dataset. In this example the deep learning techniques allowed capturing more complex patterns contained in the transaction

data that are usually hidden to traditional methodologies. The application of SHAP analysis allowed us to gain additional clarity with respect to feature importance — transaction amount and time stood out as some of the most powerful indicators of fraud. This transparency improved the inter-pretability of the model, and also created trust among stakeholders who could now get visibility into the reasons for each one of his predictions. This capability, as well as the utilization of synthetic data for augmenting supervised training, contributed to its adaptability under real-world scenarios. In summary, the performance results show that this comprehensive method can not only enhance detection rates of fraudulent cases but may also provide a solution for higher resilience and more transparent financial security systems.

5. Conclusion

In this paper, we proposed a system that combines deep learning techniques with SHAP analysis and built evaluation tools to monitor and assess the process of predicting fraud as well. With a high level of accuracy and inter-pretability, the model shows promise for various real-world applications like using as a tool to lessen the risk associated with digital transactions. Nevertheless, in the future, improvements are available that can make the system even more effective. A potential way could be to go in the direction of a federated learning concept, which learns and adapts from decentralized data sources without compromising user privacy. Finally, feeding in real-world datasets and frequent updating of the model with new transaction patterns to adapt to emerging fraud strategies for better results. Extra work with complicated models like RNNs and attention mechanisms for better capturing transaction sequences over time. In addition, one can investigate how to seamlessly integrate multi-modal data sources such as behavioral and contextual information that would together provide a complete understanding of possible fraud scenarios and result in more accurate predictions leading to sustained resilience in the fraud detection system.

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